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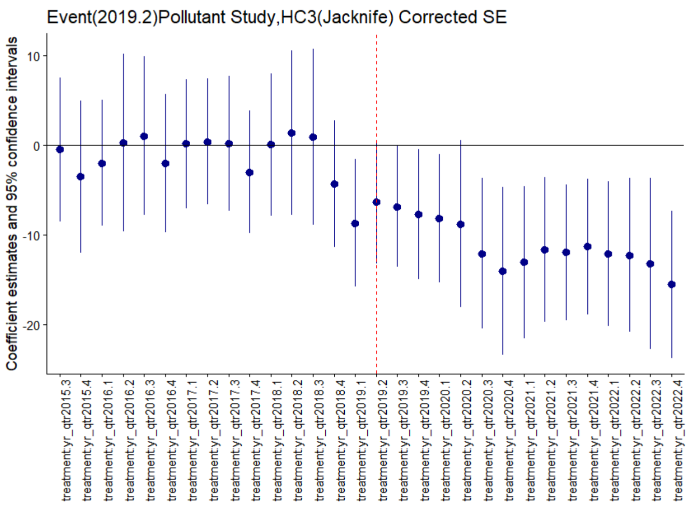
title: “An information theoretic workthrough of the Parallel trends assumption”

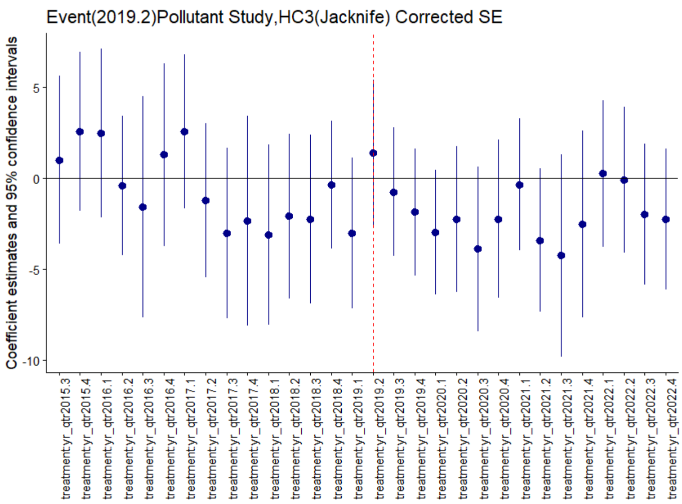
output: html_document

date: “2026-07-17”

Motivation

The DiD setup in the main ULEZ paper relies explicitly on the parallel trends assumption for constructing valid counterfactuals and estimating causal effects in the presence of selection bias. As was noted in the specification testing and event study section there’s evidence for the assumption holding mostly for PM10 but failing in the strong form for NO2. Specifically the visual and modified restriction test evidence points to some form of anticipation effect and not long term drift. The goal of this exercise is to use relative likelihoods and evidence ratios to triangulate how exactly was the assumption violated by proposing a set of models with different forms of anticipation up to fully non parallel trends.





Methodology

Estimators

The original paper uses HC3 sandwich variance covarience estimators for specification testing, in order to maintain consistency and same evidence weights we are going to use them as our robust variance estimators as well(note that this is a deviation from a pure likelihood approach in TIC and MBIC). We’d incorporate these adjustments via generalized AIC and BIC alternatives. Takeuchi and his criterion is essentially a generalisation of AIC under s model misspecification, instead of asssuming a well specified model under which the bias adjustment to relative likelihood simplifies to 2*number of free parameters.In line with Takeuchis approach we allow the bread and meat of vcov matrix to be unequal and thus use the full trace penalty tr(inv(H)%_%J) where J is the empirical score matrix. Recall that the HC3 vcov estimator is essentially inv(H)%%Omega(HC3)%%inv(H) or B%%M%_%B. Than we can derive a Takeuchi style penalty by multiplying the left side with the inverse of the inverse of the Hessian leading to

TIChc3=-2*logLik(M)+ 2*tr(_Omega(HC3)%%inv(H))(by commutivity of the trace)_

```
tic_glm <- function(model) {
  ll <- logLik(model)

  # Extract the standard and robust variance-covariance matrices
  H_inv <- vcov(model)
  V_robust <- sandwich::vcovHC(model,type="HC3")

  # Find matching parameters present in both matrices
  common_names <- intersect(rownames(H_inv), rownames(V_robust))

  # Subset both to ensure they are perfectly aligned
  H_inv_sub <- H_inv[common_names, common_names]
  V_robust_sub <- V_robust[common_names, common_names]

  # Alternative to solve(A, B): use %*% directly with H_inv since it is already inverted (H^-
1)
  # The trace penalty is trace(H * V_robust). Since vcov() is H^-1, we invert it back:
  H <- solve(H_inv_sub)
  penalty_trace <- sum(diag(H %*% V_robust_sub))

  as.numeric(-2 * ll + 2 * penalty_trace)
}
```

Where the trace penalty are effective degrees of freedom of the model. Misspecified Quasi BIC can be derived analogously by replacing the 2 with log(n) to maintain consistency at larger sample sizes:

MBIChc3=-2*logLik(M)+ log(n)*tr(_Omega(HC3)%%inv(H))_

```r
mbic_glm <- function(model) {
  ll <- logLik(model)

  # Extract the standard and robust variance-covariance matrices
  H_inv <- vcov(model)
  V_robust <- sandwich::vcovHC(model,type="HC3")

  # Find matching parameters present in both matrices
  common_names <- intersect(rownames(H_inv), rownames(V_robust))

  # Subset both to ensure they are perfectly aligned
  H_inv_sub <- H_inv[common_names, common_names]
  V_robust_sub <- V_robust[common_names, common_names]

  # Alternative to solve(A, B): use %*% directly with H_inv since it is already inverted (H^-1)
  # The trace penalty is trace(H * V_robust). Since vcov() is H^-1, we invert it back:
  H <- solve(H_inv_sub)
  penalty_trace <- sum(diag(H %*% V_robust_sub))

  as.numeric(-2 * ll + log(nobs(model)) * penalty_trace)
}
```

When leverage of residuals is low and evenly distributed corresponding to low misspecification the two penalties are very close to AIC and BIC respectively. The two estimators however are robust against heteroskadasticity and retain ability to consistently weight evidence between models with differently misspecified variences.

Using Information Criteria for (Relative) Likelihoods inference

TIC(GIC framework)

Takeuchis criterion as the generalisation of AIC is the bias adjusted metric for relative information loss between a set of models that occurs when we try and approximate a complex Data Generating process with simplified models estimated from sample models via maximum likelihood. Given that maximised log likelihoods of a set of models is a measure of how probable(or close) data is given a model and penalty term corrects the upward bias of sample estimators its possible to use differences in TICs between two models to estimate Relative Likelihoods that one model is the better approximation to DGP in the following way

RelativeLikelihood(M2 to Mmin(TIC))= exp(-0.5(changetic(M2 to Mmin(TIC))))

Where the change is calculated with respect to the model with the lowest TIC. After we calculate Relative likelihoods for all models in the candidate set we can calculate the Takeuchi/Akaike weight which corresponds to the probability that a given model in the candidate set minimizes information loss/is the best approximation to DGP, and is a ratio of Relative likelihood of Model to lowest TIC model to all other Relative Likelihoods:

TakeuchiAkaikeWeight(Mi)= RLi(Mi)/sum over all M (RLi).

This approach allows one to obtain a full probability distribution of evidence for being the best predictive/approximating model over the entire set of candidate models. The approach is essentially a Relative(to dimension given by parameters) Likelihood ratio approach to inference. Notably the model selected is the best predictive(not necessarily correct) model as we explicitly assume that the true model is too complex to be in the candidate set.This means that with large sample sizes the criterion is more than happy to pick more complex models as long as the signal to noise trade off is favorable and is thus not an asymptotically consistent selection method when the true model is suspected to be in the candidate set. Note also that we are using a sandwich based approximation instead of the true TIC to be in line with the original paper

MBIC(GBIC framework)

The log(n) adjustment can be added to the TIC to put it in line with the work of Schwarz on approximate Bayesian model selection.

MBIChc3=-2*logLik(M)+ log(n)*tr(*Omega(HC3)%%inv(H))*

Under the assumption that the true model is in the candidate state and that all models are equally likely prior to observing any data(unit prior), the relative likelihood derived from MBIC is a large sample approximation to the Bayes Factor of two models:

RelativeLikelihood(M2 to Mmin(MBIC))= exp(-0.5(changembic(M2 to Mmin(MBIC))))=BF2toMminmbic

Given that the Bayes factors is a ratio of posterior probabilities after observing data for same priors, its possible to calculate the posterior probability of a given model given data also known as a Schwarz Weight as:

SchwarzWeight=P(Mi|Data)=BFmi/sum over all M(BFmi/mlowestbic).

Therefore under the assumption that the true model is in the candidate set and unit priors one can reasonably use Schwarzs approximation with MBIC to make probability statements about plausibility of different models given data, its the most direct substitute for NHST for our purposes. Still these are quasi-posteriors and should be interpreted as plausibility weights at best

Multimodel approach to evaluating the parallel trends assumption

What we want to find out is the degree to which the assumption is violated. Given that we use quarterly means this implies that we could have no anticipation, an anticipation of 1-3 quarters, an anticipation of a year, or multi year anticipation which logically implies that the trends diverged long before policy.

For multimodel inference this means restricting the policy effect coefficients to just post policy period, and then estimating models with extra anticipation coefficients for each of 1-4 models, and using the full unrestricted event study model as a stand in for fully non parallel trends. Given that we have a model for each of the possible combinations of the trend regimes it is plausible that the true model is in our candidate set, Schwarz weights are therefore appropriate. Given that we are assuming that each of the regimes is equally likely without observing the data, we are essentially letting the data speak for itself about the relative posterior evidence weight and model uncertainty on a continous scale. This is the key advantage as we can avoid the Lindley paradox of “accepting the null” of no significant pre trend violations and actually see the embedded uncertainty of our in sample evidence.

Finally, the value of this exercise lies not just in being able to transparently evaluate evidence for effect identification, but also in doing a probabilistic post mortem of sorts on the exact direction and size of bias in our effect estimates for both PM10 and NO2 given that the original paper used the models assuming clean parallel trends. Furthermore, we can also explore how driver behavior adjusted dynamically long before policy implementation.

Estimation

We estimate all the models in the candidate set and calculate their TIC scores:

```
no2quarterly <- FullDataset %>%
  mutate(yr_qtr = quarter(date, with_year = TRUE)) %>%
  group_by(yr_qtr, Station) %>%
  summarise(mean_value = mean(no2, na.rm = TRUE), .groups = "drop") %>%
  arrange(Station, yr_qtr) %>%
  group_by(Station) %>%
  mutate(
    pct_changepol = (mean_value - lag(mean_value)) / lag(mean_value) * 100
  ) %>%
  ungroup()
#Learnig how  to use regex was a bit of a pain#
no2quarterly <- no2quarterly %>%
  mutate(City = case_when(
    str_detect(Station, regex("Birmingham", ignore_case = TRUE)) ~ "Birmingham",
    str_detect(Station, regex("Glasgow", ignore_case = TRUE))    ~ "Glasgow",
    str_detect(Station, regex("Leeds", ignore_case = TRUE))      ~ "Leeds",
    str_detect(Station, regex("London", ignore_case = TRUE))     ~ "London",
    str_detect(Station, regex("Manchester", ignore_case = TRUE)) ~ "Manchester",
    str_detect(Station, regex("Newcastle", ignore_case = TRUE))  ~ "Newcastle",
    str_detect(Station, regex("Sheffield", ignore_case = TRUE))  ~ "Sheffield",
    str_detect(Station, regex("Camden", ignore_case = TRUE))     ~ "London",
    str_detect(Station, regex("Salford", ignore_case = TRUE))    ~ "Salford",
    str_detect(Station, regex("Sunderland", ignore_case = TRUE)) ~ "Sunderland",
    str_detect(Station, regex("Westminster", ignore_case = TRUE)) ~ "London",
    TRUE ~ "Other" # Catch-all for anything missed
  ))
#Defining groups and Intervention Period#
intervention <- c(2019.2,2019.3,2019.4,2020.1,2020.2,2020.3,2020.4,2021.1,2021.2,2021.3,2021.
4,2022.1,2022.2,2022.3,2022.4)
treatment <- c(
  "camden-- euston road",
  "london-bloomsbury",
  "LondonC",
  "LondonMR",
  "LondonW"
)

control <- c(
  "glasgow-anderston",
  "glasgow-byres road",
  "glasgow-high street",
  "glasgow-townhead",
  "GlasgowK",
  "manchester-oxford road",
  "ManchesterPic",
  "NewcastleC",
  "salford-eccles",
  "salford-m60"
)
panel_dfind <- no2quarterly %>%
  mutate(
    intervention = ifelse(yr_qtr %in% intervention, 1,0),
    control = ifelse(Station %in% control, 1, 0),
    treatment= ifelse(Station %in% treatment,1,0),
    did= intervention*treatment)
```

```
treatment_control_df <- panel_dfind %>%
  filter(treatment + control == 1,yr_qtr < 2023.1 & yr_qtr > 2015.1)
#Estimating Models#
treatment_control_df$Station <- as.factor(treatment_control_df$Station)
treatment_control_df$yr_qtr <- as.factor(treatment_control_df$yr_qtr)
treatment_control_df$City <- as.factor(treatment_control_df$City)
treatment_control_df$did <-as.factor(treatment_control_df$did)
reducedmodel <- lm(mean_value~did+yr_qtr+Station,data=treatment_control_df)
treatment_control_df$ulez_id <- ifelse(treatment_control_df$did == 1,
                                       as.character(treatment_control_df$Station),
                                       "Control")
hetmodel <- lm(mean_value~did+did:Station+yr_qtr+Station,data=treatment_control_df)
eventmodel <- lm(mean_value~treatment*yr_qtr+Station,data=treatment_control_df)
post_quarters_null <- c(
  "2019.2", "2019.3", "2019.4",
  "2020.1", "2020.2", "2020.3", "2020.4",
  "2021.1", "2021.2", "2021.3", "2021.4",
  "2022.1", "2022.2", "2022.3", "2022.4"
)

post_quarters_ant1 <- c("2019.1", post_quarters_null)
post_quarters_ant2 <- c("2018.4", post_quarters_ant1)
post_quarters_ant3<- c("2018.3", post_quarters_ant2)
post_quarters_ant4<- c("2018.2", post_quarters_ant3)
# 2. Build explicit interaction terms in the dataframe
# This creates a factor that behaves like yr_qtr for target quarters,
# but groups all other quarters into a single "Pre-Treatment/Baseline" reference category (0
effect).
prepared_df <- treatment_control_df %>%
  mutate(
    # Convert factor to character inline to preserve text values
    yr_qtr_char = as.character(yr_qtr),

    # 1. Corrected DiD term using multiplication
    did = intervention * treatment,

    # 2. Preserved quarter strings in the interaction terms
    yr_qtr_null = if_else(yr_qtr_char %in% post_quarters_null, yr_qtr_char, "Pre-Treatment"),
    yr_qtr_ant1 = if_else(yr_qtr_char %in% c("2019.1"), yr_qtr_char, "Pre-Treatment"),
    yr_qtr_ant2 = if_else(yr_qtr_char %in% c("2018.4", "2019.1"), yr_qtr_char, "Pre-Treatmen
t"),
    yr_qtr_ant3 = if_else(yr_qtr_char %in% c("2018.3", "2018.4", "2019.1"), yr_qtr_char, "Pre
-Treatment"),
    yr_qtr_ant4 = if_else(yr_qtr_char %in% c("2018.2", "2018.3", "2018.4", "2019.1"), yr_qtr_
char, "Pre-Treatment"),
    yr_qtr_unrestrict=ifelse(!yr_qtr %in% post_quarters_null,yr_qtr_char, "Pre-Treatment"))


# 3. Explicitly estimate the restricted models
# We interact 'treatment' ONLY with our newly capped quarter variables.
# Note: Use treatment : yr_qtr_xxx to get the interaction effects without changing the main e
ffects.

# Baseline / Unrestricted Model
eventmodelunrestrict <- lm(mean_value ~ did+yr_qtr_unrestrict:treatment+ Station+yr_qtr, data
= prepared_df)
```

```
unrestrictvcov <- vcovHC(eventmodelunrestrict,type = "HC3")
# Restricted Model 1: Null (No Anticipation)
eventmodelnull <- lm(mean_value ~  did+Station + yr_qtr, data = prepared_df)
nullvcov <- vcovHC(eventmodelnull,type = "HC3")
# Restricted Model 2: 1-Quarter Anticipation
eventmodelant1 <- lm(mean_value ~ did+treatment : yr_qtr_ant1 + Station + yr_qtr, data = prep
ared_df)
ant1vcov <- vcovHC(eventmodelant1,type = "HC3")
# Restricted Model 3: 2-Quarter Anticipation
eventmodelant2 <- lm(mean_value ~ did+treatment : yr_qtr_ant2 + Station + yr_qtr, data = prep
ared_df)
ant2vcov <- vcovHC(eventmodelant2,type = "HC3")
eventmodelant3 <-  lm(mean_value ~ did+treatment : yr_qtr_ant3 +Station + yr_qtr, data = prep
ared_df)
ant3vcov <- vcovHC(eventmodelant3,type = "HC3")
eventmodelant4 <-  lm(mean_value ~ did+treatment : yr_qtr_ant4 + Station + yr_qtr, data = pre
pared_df)
ant4vcov <- vcovHC(eventmodelant4,type = "HC3")
model_coef_list <- list(
  unrestricted = coef(eventmodelunrestrict),
  ant4         = coef(eventmodelant4),
  ant3         = coef(eventmodelant3),
  ant2         = coef(eventmodelant2),
  ant1         = coef(eventmodelant1),
  null         = coef(eventmodelnull)
)
 vcov_list <- list(
       unrestricted = unrestrictvcov,
       ant4         = ant4vcov,
       ant3         = ant3vcov,
       ant2         = ant2vcov,
       ant1         = ant1vcov,
       null         = nullvcov)


effect_summary_tableno2 <- names(model_coef_list) %>%
    map_df(function(model_name) {
             mod  <- model_coef_list[[model_name]]
             vcov <- vcov_list[[model_name]]

               # Extract coefficient (returns NA if dropped/collinear)
               did_coef <- mod["did"]

                # Extract variance safely from the robust vcov matrix
                 did_var <- if ("did" %in% colnames(vcov)) vcov["did", "did"] else NA

                   # Calculate Standard Error for standard reporting completeness
                   did_se <- sqrt(did_var)

                   tibble(
                           Model        = model_name,
                           DiD_Estimate = did_coef,
                           Robust_Var   = did_var
                   )})
```

We also calculate their MBIC scores:

```
tablembic <- t(tibble("unrestrictedtrends"=mbic_glm(eventmodelunrestrict),
                     "1 year anticipation"=mbic_glm(eventmodelant4),
                     "3 quarters anticipation"=mbic_glm(eventmodelant3),
                     "2 quarters anticipation"=mbic_glm(eventmodelant2),
                     "1 quarter anticipation"=mbic_glm(eventmodelant1),
                     " no pre treatment effects"=mbic_glm(eventmodelnull)))
MBICscore <- c(as.numeric(t(tablembic)))
```

We can know use the TIC scores to derive and summarize their Relative Likelihoods and Takeuchi weights for a case in which we don't assume that the model is in the candidate set:

```
tabletic <- t(tibble("unrestrictedtrends"=tic_glm(eventmodelunrestrict),
                     "1 year anticipation"=tic_glm(eventmodelant4),
                     "3 quarters anticipation"=tic_glm(eventmodelant3),
                     "2 quarters anticipation"=tic_glm(eventmodelant2),
                     "1 quarter anticipation"=tic_glm(eventmodelant1),
                     " no pre treatment effects"=tic_glm(eventmodelnull)))
TICscore <- c(as.numeric(t(tabletic)))
Model<- c("unrestrictedtrends","1 year anticipation","3 quarters anticipation","2 quarters anticipation","1 quarter anticipation"," no pre treatment effects")
TICch <- TICscore - min(TICscore)
RL <- exp(-0.5*TICch)
Weight <- round(RL/sum(RL),4)
resultsno2tic <- data.frame(
  'Model(J=HC3)' = Model,
  TIC = TICscore,
  Delta = TICch,
  RelLikelihood = RL,
  Weight = Weight,
  check.names = FALSE)
```

Our main table uses the mbic to calculate schwarz posterior probability weights:

```
MBICch <- MBICscore - min(MBICscore)
BF <- exp(-0.5*MBICch)
PosteriorP <- round(BF/sum(BF),4)
resultsno2mbic <- data.frame(
  'Model(J=HC3)' = Model,
  MBIC = MBICscore,
  Delta = MBICch,
  BayesFactor = BF,
  `P(M|Data)` = PosteriorP, # Backticks allow special characters
  check.names = FALSE)
```

We know rerun the analysis for pm10 to get results for pm10 side of the analysis

• ◦ ▪

NO2

```
gt(resultsno2mbic)
```

Model(J=HC3)	MBIC	Delta	BayesFactor	P(M|Data)
unrestrictedtrends	2856.566	182.1273205	2.828491e-40	0.0000
1 year anticipation	2726.806	52.3672988	4.251925e-12	0.0000
3 quarters anticipation	2704.596	30.1565295	2.828739e-07	0.0000
2 quarters anticipation	2674.439	0.0000000	1.000000e+00	0.5280
1 quarter anticipation	2674.665	0.2254438	8.933991e-01	0.4717
no pre treatment effects	2689.748	15.3084747	4.740312e-04	0.0003

```
gt(resultsno2tic)
```

Model(J=HC3)	TIC	Delta	RelLikelihood	Weight
unrestrictedtrends	2506.016	52.187868	4.651023e-12	0.0000
1 year anticipation	2469.457	15.629173	4.038017e-04	0.0003
3 quarters anticipation	2463.176	9.348153	9.334140e-03	0.0078
2 quarters anticipation	2453.828	0.000000	1.000000e+00	0.8346
1 quarter anticipation	2457.166	3.337935	1.884415e-01	0.1573
no pre treatment effects	2475.713	21.884910	1.769100e-05	0.0000

The approximate Schwarz posterior weights predominantly favor the anticipation models with 2 or 1 quarter of anticipation being about (0.528+0.4717)/0.0003=3332 times more likely than the clean DID model. At the same time the higher order anticipation and the full event study models have relatively litlle support and are weighted near zero. Under the assumption that the true model in the set has a well specified mean we have that the 3-6 months of anticipation was the most likely driver and composition effect response to the ULEZ announcement, there's low to no sample evidence support for non parallel trends though.Still whether anticipation was 2 or 3 quarters remains fundamentally uncertain with the models having not enough posterior sample evidence to distiguish between 1 and 2 quarters with the approximate bayes factor(posterior evidence ration) of 1:1.1 in favour of the 2 quarter model.

As a robustness check we also drop the true model assumption and consider Takeuchi results as well. We are know looking for the lowest information loss/best predictions on unseen data(relative KL divergence) model. This is justified since if seasonal interactions or non linearity are ommited the model for the mean is biased and the true model isn't in the candidate set, TIC is thus more lenient towards including predictors. The weights heavily favour the 2 quarter model with it being 5.5 times more likely to be the best approximation in the set than the 1 quarter model. If we don't have the approximately true DGP assumption to allow us to use MBIC, and look for best predicitive approximation instead we have weakly strong evidence for 2 periods of anticipation effect as the best approximation, still this fully hinges on the assumtion that the true model is not in the set and we are minimising prediction error. If the approximately true model is included however(if we take our candidate set as exhaustive) than we can't distinguish between the 1 and 2 quarter anticipation model.

PM10

```
gt(resultspm10mbic)
```

Model(J=HC3)	MBIC	Delta	BayesFactor	P(M|Data)
unrestrictedtrends	2447.128	82.606344	1.154145e-18	0.0000
1 year anticipation	2380.675	16.153208	3.107245e-04	0.0002
3 quarters anticipation	2375.499	10.977479	4.133050e-03	0.0030
2 quarters anticipation	2369.113	4.591336	1.006941e-01	0.0735
1 quarter anticipation	2367.179	2.657455	2.648140e-01	0.1933
no pre treatment effects	2364.522	0.000000	1.000000e+00	0.7300

```
gt(resultspm10tic)
```

Model(J=HC3)	TIC	Delta	RelLikelihood	Weight
unrestrictedtrends	2169.023	11.8545461	0.002665741	0.0009
1 year anticipation	2159.776	2.6071060	0.271565211	0.0883
3 quarters anticipation	2158.922	1.7537533	0.416080453	0.1353
2 quarters anticipation	2157.800	0.6317395	0.729154400	0.2371
1 quarter anticipation	2157.169	0.0000000	1.000000000	0.3252
no pre treatment effects	2158.013	0.8439356	0.655755151	0.2132

The approximate posterior weights are spread out in such a way that the clean DiD model is 2.68 times more likely than the anticipation models. The noise in PM10 data is likely such that the models cannot pick up any meaningful divergence between the treatment and control under the approximately true DGP assumption. This alligns with the conclusions of the original paper - the volume effect which would have affected PM10 more through the decrease in the total number of cars on the road was relatively modest, with people acqurining cleaner vehicles but largely leaving their driving habits unchanged.

Dropping the true DGP assumption and lookign for the best approximating model instead sheds light on uncertainty in PM10 time series. Although some form of anticipation is 5 times more likely than clean DiD to be the better approximation,the evidence doesn't favor any individual model as the best approximating one. Like with NO2 and even more so we arguably don't have enough evidence to pinpoint any particular anticipation pattern or lack thereof given the sample and its noise.

Takeuchi and Akaike vs Schwarz and Bayes and GIC vs GBIC

Ultimately whether one should choose the relative information loss or posterior probability interpretation of the analysis outcome boils down to whether one thinks the mean is well specified(additive and linear) or misspespecified (which is most likely). I tend to err on the side of caution and use both and interpret both jointly even if they come from two competing philosophies. Still the relative evidence approach does help hint towards most plausible anticipation patterns in NO2 which are economically meaningful if DGP assumption holds and at least provides us with the decent enough representation of inherent in sample uncertainty if it doesn't.

Bonus:Bayesian Model Averages of the Anticipation Adjusted Effects to put a number on attentuation bias

NO2: Post-policy ATT by Model with HC3 variences

Model	ATT	VAR
unrestricted	−2.292	2.6542499
1 year anticipation	−10.261	0.7714553
3 quarters anticipation	−10.425	0.7644262
2 quarters anticipation	−10.532	0.7665021
1 quarter anticipation	−10.272	0.7311806
no pre treatment effects	−9.768	0.7239094

PM_{10}: Post-policy ATT by Model with HC3 variences

Model	ATT	VAR
unrestricted	1.2039	2.1086
1 year anticipation	−1.6237	0.4334
3 quarters anticipation	−1.4682	0.4076
2 quarters anticipation	−1.2908	0.3926
1 quarter anticipation	−1.3045	0.3532
no pre treatment effects	−1.1083	0.3300

When there are anticipation effects but no cumulative long term drift the DiD coefficient has the correct direction but has incorrect effect size as the counter factual is calculated at a lower baseline than when the policy actually started taking effect. As a nice side effect of having full probability distributions over all our theories we can try and adjust for attenuation bias by calculating a weighted average of all model estimates given their probabilities note that since the probability weight on the unrestricted model is zero it has no effect on the adjustment.

```
gt(data.frame(NO2BAME=sum(resultsno2mbic$`P(M|Data)`*no2_att$ATT),
NO2BAMEtic=sum(resultsno2tic$Weight*no2_att$ATT),
PM10BAME=sum(resultspm10mbic$`P(M|Data)`*pm10_att$ATT),
PM10BAMEtic=sum(resultspm10tic$Weight*pm10_att$ATT)))
```

NO2BAME	NO2BAMEtic	PM10BAME	PM10BAMEtic
-10.4094	-10.49054	-1.160855	-1.307499

Its notable that the weighted coefficients for NO2 and Pm10 are larger than the anticipation affected ones by about 0.574(5.9 percent loss to attenuation) and 0.053(4.61 percent loss to attenuation) respectively, this gap represents the expected attenuation bias for our model set(note that getting SE for BMAs is notoriously difficult so these are just descriptive point estimates,though we make an effort below).Estimated Standard errors incorporating both within and between model noise however are larger than the BMA adjustments, implying that the changes are indistiguishible from noise.Relaxing the assumption that the true DGP is in candidate set and using takeuchi weights increases the expected bias. The original choice to stick with original policy timings therefore was more or less robust to anticipation bias for both pollutants for a given sample size Given BMA noise as even in the most pessimistic outcome the uncertainty doesn't seem to overturn the original specification.

```
# 1. Compute the Point Estimates first
no2_bma_m  <- sum(resultsno2mbic$`P(M|Data)` * no2_att$ATT);
no2_bma_t  <- sum(resultsno2tic$Weight * no2_att$ATT);
pm10_bma_m <- sum(resultspm10mbic$`P(M|Data)` * pm10_att$ATT);
pm10_bma_t <- sum(resultspm10tic$Weight * pm10_att$ATT);

# 2. Compute the Standard Errors using the Law of Total Variance
no2_se_m  <- sqrt(sum(resultsno2mbic$`P(M|Data)` * (no2_att$VAR + (no2_att$ATT - no2_bma_m)^
2)));
no2_se_t  <- sqrt(sum(resultsno2tic$Weight * (no2_att$VAR + (no2_att$ATT - no2_bma_t)^2)));
pm10_se_m <- sqrt(sum(resultspm10mbic$`P(M|Data)` * (pm10_att$VAR + (pm10_att$ATT - pm10_bma_
m)^2)));
pm10_se_t <- sqrt(sum(resultspm10tic$Weight * (pm10_att$VAR + (pm10_att$ATT - pm10_bma_t)^
2)));

# 3. Present cleanly side-by-side using gt
gt(data.frame(
  Metric      = c("BMA Estimate", "Standard Error"),
  NO2_MBIC    = c(no2_bma_m, no2_se_m),
  NO2_TIC     = c(no2_bma_t, no2_se_t),
  PM10_MBIC   = c(pm10_bma_m, pm10_se_m),
  PM10_TIC    = c(pm10_bma_t, pm10_se_t)
))
```

Metric	NO2_MBIC	NO2_TIC	PM10_MBIC	PM10_TIC
BMA Estimate	-10.4094016	-10.4905413	-1.1608548	-1.3074992
Standard Error	0.8756876	0.8774664	0.5889994	0.6325832